

Student Takeaway for Sync Session

<<Week 14: Model Regularisation and Evaluation>>

This guide is your roadmap to making the most of our online session. Packed with essential tips and strategies, it's designed to keep you engaged, prepared, and ready to dive into a smooth and productive learning journey. Get ready to participate, learn, and thrive!

Session Overview

Session title	Model Regularisation and Evaluation		
Session duration	1 hour 45 minutes		
Session type	• Lectures: Understanding regression models, challenges of underfitting/overfitting, role of regularisation, evaluation metrics. • Case Studies & Code: Boston Housing dataset, Ridge and Lasso regression implementation.		
Scope	This session covers model regularisation and evaluation in predictive modeling. Learners will explore: • Linear, multiple, and polynomial regression • Underfitting vs. overfitting • Regularisation (Ridge and Lasso) with practical examples • Evaluation metrics: R^2, MAE, MSE, RMSE		
Learning objectives			
	Objective		Core capability
	Understand regression model types		Foundational modelling skills
	Identify overfitting and underfitting		Diagnostic awareness
	Apply Ridge and Lasso regularisation		Controlling model complexity
	Evaluate models using key metrics		Model validation and interpretation
Software/Tools	• Python (Libraries: scikit-learn, numpy, matplotlib, pandas) • IDE: Jupyter Notebook • Dataset: Boston Housing Dataset • Presentation Tool: PowerPoint		

Pro tips for success:

Engage Actively: Ask why a model might overfit or underfit.

Practice the Code: Re-run Ridge/Lasso with different alpha values.

Reflect on Results: Visualise actual vs. predicted to check accuracy.

Session Details

Slide(s)	Topic/Issue	What to expect	Notes
1-2	Model Regularisation and Evaluation	Goals and structure overview	Focus on balance in modelling
3	Regression Types	Linear, multiple, polynomial	Basis for all predictive modelling
4	Model Fit	Underfitting, overfitting, good fit	Use cases and visual illustrations
5	Regularisation Regularisation Benefits	Why it matters in complex models Generalisation and prediction reliability	Prevents learning noise Better long-term performance
6	Recap	Summary of motivation behind regularisation	Sets stage for implementation
7-9	Boston Housing Dataset	Real-life dataset for practice	Predicting house prices
10-19	Regularisation in Regression using Ridge and Lasso	- Load housing data - Use polynomial regression - Apply Ridge & Lasso regularisation - Compare R^2 and RMSE Visualise actual vs predicted prices	- Load data - Use Area - Train/test split - Poly + Ridge/Lasso - Evaluate R^2 & RMSE Plot results
20-21	Why Evaluation Matters	Training vs real-world performance	Importance of test data results
22	Metrics Overview	R^2 , MAE, MSE, RMSE	Learn what each metric tells you
23	Understanding R^2 Score	Higher R^2 = better fit, less scatter	High $R^2 \neq$ always good (check overfitting). Negative = worse than baseline.
24	Metric Implementation	Code snippet with results	Evaluate predictions quantitatively
25	Train vs Test Gap	Overfitting signs	Compare scores to validate model

Slide(s)	Topic/Issue	What to expect	Notes
26	Actual vs Predicted	Visualise model accuracy	Look for clustering on $y=\hat{y}$ line
27	Key Takeaways	Regularisation + evaluation = reliability	Metric choice impacts decisions
28	Wrap-Up & Q&A	Review session + open floor	Ask your model questions here

Post session

Reflection challenge	What did you learn about model evaluation that changes how you'll approach future ML projects?
Explore more	• Read: Articles on Ridge vs. Lasso pros and cons • Watch: Visual demos on underfitting vs. overfitting • Experiment: Change alpha values in Ridge and see the impact on R^2
Get inspired	The Boston Housing model isn't just academic—it mirrors real estate and urban planning needs. How might you apply regularisation in your domain?
The journey ahead	Get ready for the next session on Model Interpretability, where you'll learn to explain your model's decisions clearly and confidently.